SIMATS ENGINEERING

ASSESSMENT TOOL 3: Game Based Learning Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 40 %

Total Marks: 50 **Time: 60 Mins**

No. of Questions: 5

Annexure A

Game Based Learning Assessment

Code of Conduct:

I Kamalli shree V(192525030) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Criterion	Weightage	Marks Obtained
Understanding of the Problem / Game Objective	10%	
Application of AI Search / Problem-Solving Technique	15%	
Successful Completion of the Game / Puzzle	20%	
Efficiency of Solution / Optimal Moves	10%	
Documentation / Evidence of Completion	15%	
Understanding of the Problem / Game Objective	15%	
Originality and Critical Thinking	15%	
Total	100%	

Signature of the Faculty Total Marks Awarded

Answer All the Questions

Q. No.	Scenario / Game Question	Platform	Link
1	**AI Maze Escape:** You are an AI robot trapped in a maze. Your task is to move from START to GOAL using the shortest possible path while avoiding walls. **Challenge:** Find the shortest path and record the number of steps taken.	Scratch	https://scratch.mit.edu
2	**N-Queens Challenge:** You are an AI chess strategist. **Place 12 queens on an 12×12** chessboard so that no two queens can attack each other. **Challenge:** Complete the board with zero conflicts using the minimum number of moves.	Online N-Queens Game	https://www.n-queens.com
3	**Water Jug Puzzle:** You are an AI agent solving a water jug problem. You have two jugs with capacities of **11 litres and 9 litres**. Neither jug has measurement markings. **Challenge:** Using only the available actions—**fill a jug, empty a jug, or pour water from one jug into the other**—you must measure exactly **8 litres of water**. **Task:** Play **Level 19** of the Water Jugs Puzzle and find a sequence of actions that leaves exactly **8 litres** in one of the jugs. Try to solve the puzzle using the **minimum number of moves**.	Tic-Tac-Toe	https://playtictactoe.org
4	**Connect Four AI Challenge:** You are playing against an AI opponent. Your goal is to connect four of your pieces in a row before the AI does. **Challenge:** Plan your moves carefully and defeat the AI opponent.	Connect Four	https://www.mathsisfun.com/games/connect4.html

5	**8-Puzzle AI Challenge:** You are given a scrambled 8-puzzle. Move the tiles one at a time and arrange them in the correct order. **Challenge:** Solve the puzzle using the minimum possible number of moves.	8-Puzzle Game	https://8puzzle.org/

Structure of the Report – Game-Based Learning

S.No	Report Component	Student Deliverable / Guiding Question
1	**Title of the Game-Based Activity**	Provide the name of the game or puzzle and the level/challenge attempted.
2	**Game Platform / Link**	Mention the platform or website used and provide the corresponding game link.
3	**Objective of the Activity**	Explain the objective of the game and the target that needs to be achieved.
4	**Problem Statement**	Describe the problem or challenge given in the game. What is the student required to solve?
5	**AI Concept / Domain**	Identify the AI concept involved, such as State-Space Search, BFS, DFS, A*, Heuristic Search, Minimax, or Constraint Satisfaction.
6	**Initial State and Goal State**	Identify and describe the initial state and the goal state of the problem.
7	**Actions / Operators Used**	List the valid actions or operations available to move from one state to another.
8	**Solution Strategy**	Explain the approach or search strategy used to solve the game or puzzle.
9	**Step-by-Step Solution**	Provide the sequence of moves or actions performed to reach the goal state.
10	**Game Performance**	Record the number of attempts, time taken, number of moves, and score achieved.
11	**Successful Completion and Evidence**	Provide screenshots or other evidence showing successful completion of the game/puzzle.
12	**Efficiency and Optimality**	Analyse whether the solution was optimal or efficient. Can the puzzle be solved using fewer moves or less time?
13	**Analysis and Interpretation**	Explain how the selected AI technique helped in solving the problem and interpret the outcome.
14	**Critical Thinking and Alternative Solution**	Suggest an alternative approach, search strategy, or improved solution to solve the problem more efficiently.
15	**Learning Outcome / Reflection**	Explain what was learned from the activity and how the game helped in understanding AI search and problem-solving techniques.
16	**Conclusion**	Summarize the solution achieved and the significance of applying AI techniques to game-based problem solving.
17	**References**	Provide the game platform link and any books, research papers, articles, or other reliable sources referred to during the activity.

Evaluation Rubrics

Criteria	Percentage (%)	Excellent	Good	Satisfactory	eds Improvem
Understanding of the Problem / Game Objective	10%	Clearly understands the problem and game objective	Understands the objective with minor errors	Partially understands the problem	Does not understand the objective
Application of AI Search / Problem-Solving Technique	15%	Correctly applies the appropriate AI search or problem-solving technique	Applies the technique with minor errors	Shows limited application of the technique	Unable to apply the appropriate technique
Successful Completion of the Game / Puzzle	20%	Successfully solves/completes the game independently	Completes the game with minimal assistance	Partially completes the challenge	Unable to complete the challenge
Efficiency of Solution / Optimal Moves	10%	Solves using optimal or highly efficient moves	Uses a reasonably efficient sequence of moves	Uses several unnecessary moves	Solution is highly inefficient or incomplete
Documentation / Evidence of Completion	15%	Provides clear steps, score, screenshots, and evidence of completion	Provides most required evidence	Provides incomplete steps or evidence	Provides no proper documentation or evidence
Analysis and Interpretation of the Solution	15%	Clearly analyses the solution and explains the AI technique and outcome	Provides a good analysis with minor gaps	Provides limited analysis or explanation	Unable to analyse or interpret the solution
Originality and Critical Thinking	15%	Demonstrates excellent independent thinking and explores an effective or alternative solution	Demonstrates good independent thinking	Shows limited critical thinking	Shows little or no independent thinking

GAME 1 – AI Maze Escape

Title : AI Maze Escape

Platform: Scratch

Objective:

Guide the AI robot from the start position to the goal using the shortest path while avoiding obstacles.

Problem Statement:

The robot must move through the maze by selecting the correct path to reach the destination with minimum steps.

AI Concept:

- State Space Search
- Breadth First Search (BFS)

Initial State: Robot starts at Box 2.

Goal State : Robot reaches the exit.

Solution Strategy:

- Explore valid paths.
- Select the shortest route.
- Avoid unnecessary movements.

Performance:

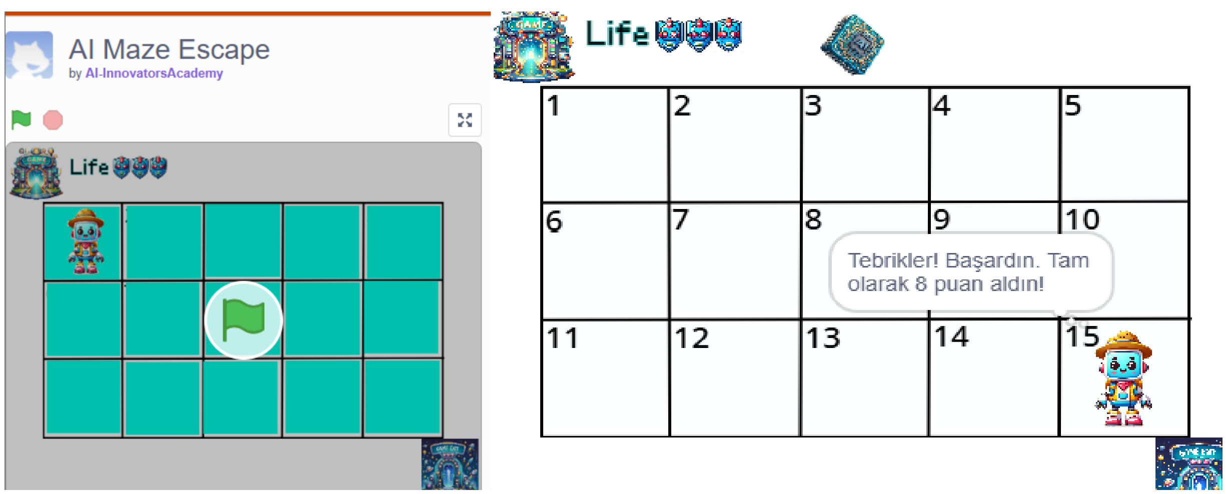
- Attempts: 1
- Moves: 5
- Result: Successfully Completed

Learning Outcome:

Learned how AI search algorithms solve pathfinding problems efficiently.

Conclusion:

The maze was solved successfully using an efficient search strategy.



GAME 2 – 12 Queens Challenge

Title: 12 Queens Challenge

Platform: Online 12×12 Queens Puzzle

Objective:

Place 12 queens on a 12×12 board so that no two queens attack each other.

Problem Statement:

Arrange queens without sharing the same row, column, or diagonal.

AI Concept:

- Constraint Satisfaction Problem (CSP)
- Backtracking Search

Initial State: Empty 12×12 chessboard.

Goal State: 12 queens placed with zero conflicts.

Solution Strategy:

- Place queens row by row.
- Check conflicts before placing each queen.
- Backtrack whenever conflicts occur.

Performance:

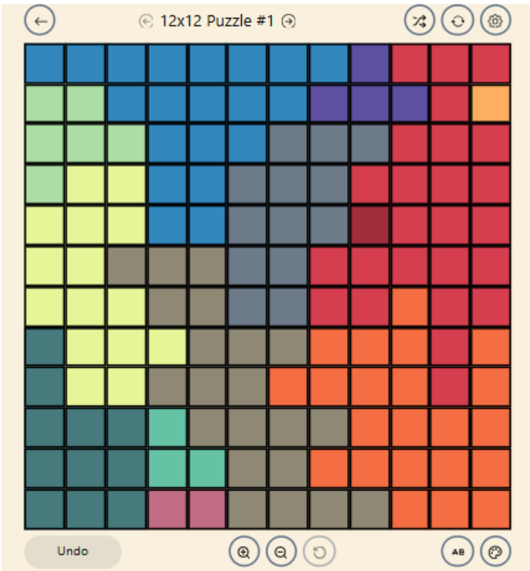
- Attempts: 1
- Result: Puzzle Solved

Learning Outcome:

Understood constraint satisfaction and backtracking techniques.

Conclusion:

The puzzle demonstrated how AI solves combinatorial optimization problems.



GAME 3 – Water Jug Puzzle

Title: Water Jug Puzzle

Platform: Water Jug Puzzle (Level 19)

Objective:

Measure exactly 8 litres using 11L and 9L jugs.

Problem Statement:

Use only filling, emptying, and pouring operations to obtain exactly 8 litres.

AI Concept

- State Space Search
- Breadth First Search (BFS)

Initial State: Both jugs are empty.

Goal State: One jug contains exactly 8 litres.

Solution Strategy:

- Fill and pour water systematically.
- Track each state.
- Reach the goal using minimum moves.

Performance:

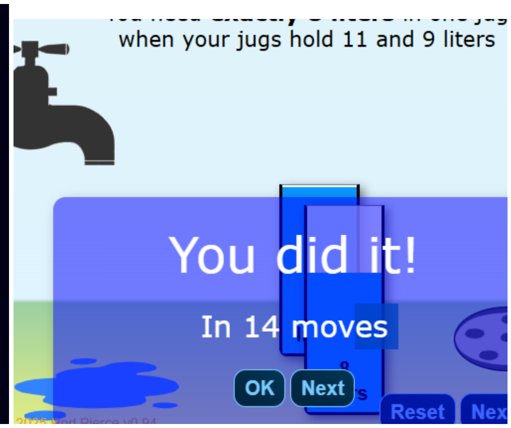
- Attempts: 1
- Result: Successfully Completed

Learning Outcome:

Learned how AI represents and explores state-space problems.

Conclusion:

The puzzle demonstrated efficient state-space search techniques.



GAME 4 – Connect Four AI Challenge

Title: Connect Four AI Challenge

Platform: Connect Four

Objective:

Connect four pieces in a row before the AI opponent.

Problem Statement:

Play strategically against an AI by creating four connected pieces while blocking the AI's moves.

AI Concept:

- Game Playing
- Minimax Algorithm

Initial State: Empty game board.

Goal State: Player connects four pieces first.

Solution Strategy:

- Control the center columns.
- Block the AI's winning moves.
- Create multiple winning opportunities.

Performance:

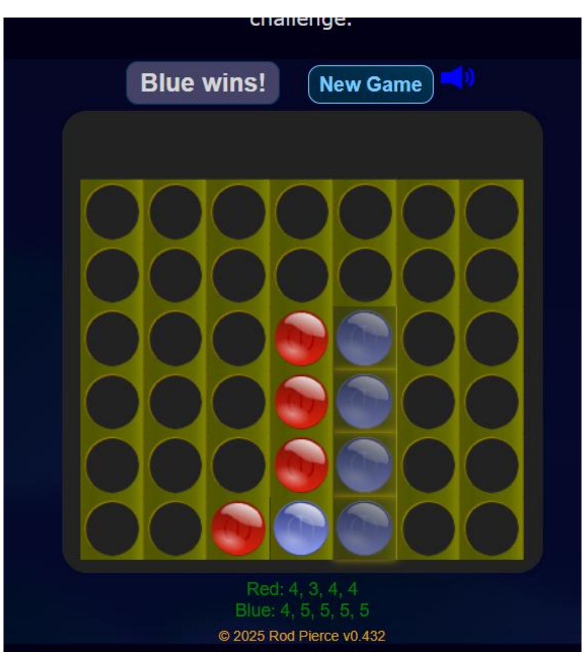
- Attempts: 1
- Result: AI Defeated

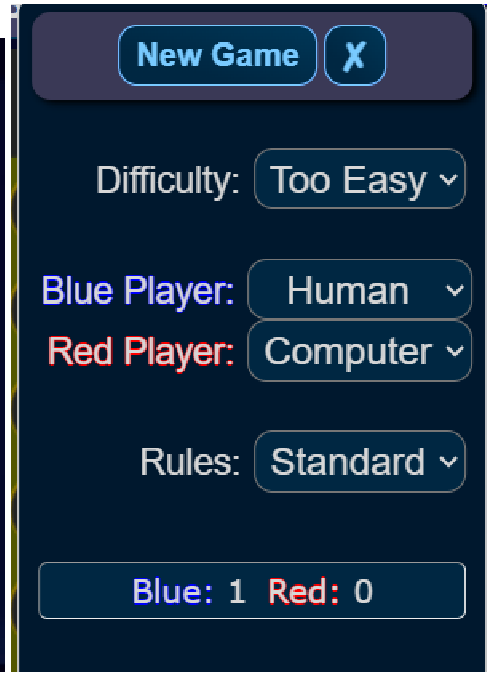
Learning Outcome:

Understood AI decision-making and game-search strategies.

Conclusion:

The activity demonstrated strategic planning using Minimax.





GAME 5 – 8 Puzzle AI Challenge

Title: 8 Puzzle AI Challenge

Platform: 8 Puzzle Game

Objective:

Arrange all numbered tiles in the correct order using the minimum number of moves.

Problem Statement:

Move one tile at a time until the puzzle reaches the goal configuration.

AI Concept:

- A* Search Algorithm
- Heuristic Search

Initial State: Scrambled tile arrangement.

Goal State: Tiles arranged in numerical order with the blank space in the final position.

Solution Strategy:

- Move only adjacent tiles.
- Use heuristic values to choose the best move.
- Minimize unnecessary movements.

Performance:

- Attempts: 1
- Result: Puzzle Solved

Learning Outcome:

Learned how heuristic search improves problem-solving efficiency.

Conclusion:

The activity demonstrated how AI uses informed search algorithms to solve optimization problems.